

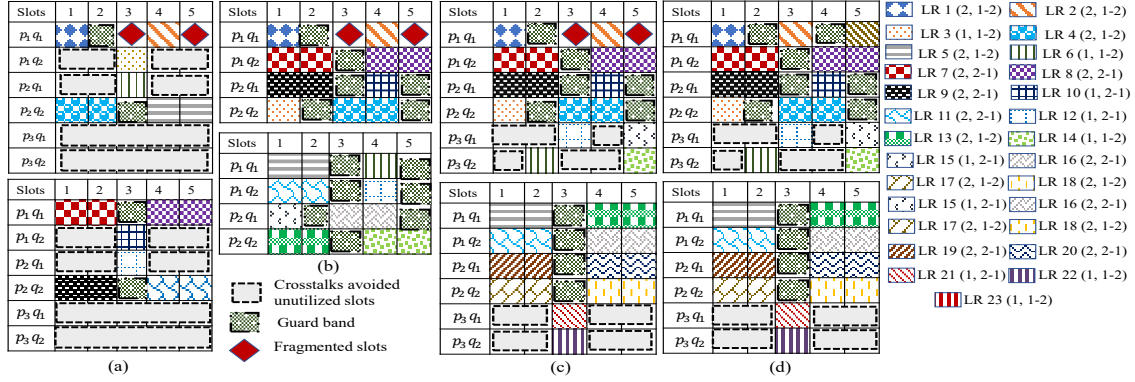


Fig. 3: Resource allocation under (a) co-propagation (no defragmentation), (b) counter-propagation with vertex elimination (no defragmentation), (c) counter-propagation with non-vertex elimination (no defragmentation), and (d) counter-propagation with non-vertex elimination and defragmentation.

number of cores and modes, and each mode supports an equal number of spectrum slots. Lightpath requests are established by satisfying IC-XT, IM-XT, and spectrum contiguity constraints.

B. Problem definition

DAAM-CP investigates resource allocation in an optical network where each node pair is connected by two fiber links, with signals propagating in a counter-propagation manner. Each link contains multiple cores, each with several modes and spectrum slots. DAAM-CP consists of two main phases: (i) Phase 1 using optimization problem to identify two disjoint sets of core-mode pairs that minimize XT under counter-propagation; (ii) Phase 2 develops an exact CTMC-based analytical model that evaluates BP while accounting for spectrum contiguity, IC-XT, IM-XT, and defragmentation, to enhance resource utilization.

IV. DAAM-CP: PROPOSED MODEL CONSIDERING NON-VERTEX ELIMINATION AND DEFRAGMENTATION

This section introduces the proposed DAAM-CP model, which operates in two phases. Phase 1 identifies two disjoint sets of core-mode pairs that minimize XT under counter-propagation; these are used as input for Phase 2. Phase 2 develops an exact CTMC-based analytical model for a link to evaluate BP, considering spectrum contiguity, IC-XT, IM-XT, and defragmentation to improve resource utilization. The overall concept of DAAM-CP is illustrated in Fig. 4. Details of both phases are provided in the following subsections.

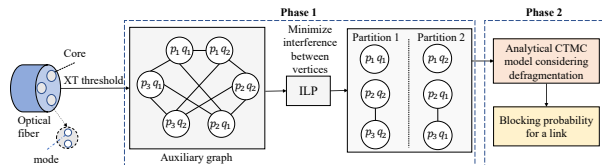


Fig. 4: Concept of DAAM-CP.

A. Phase 1: Resource identification using ILP

Inspired by the route partitioning work in [34], we define the optimization problem as a resource partitioning (RP) problem. The RP problem is formulated as an ILP problem to minimize total interference (XT) between vertices within the same partitions after bi-partitioning resources.

We define the parameters and decision variables for the RP problem as follows. The interference weight between vertices x and y , denoted as t_{xy} , is calculated based on XT. To assign these weights, we consider the relative importance or influence of each pair of vertices. This is achieved by assigning numerical values of XT ($\mathbb{C}_{\text{Tot}}^{Yr, I, d}$) between each pair of vertices. The weights are normalized to sum up to 1 (or 100%) and are assigned using a ratio scale within the range $[0, 1]$. The decision variable d_{xy} is 1 if vertices x and y are on different sides of the cut, and 0 otherwise.

We formulate the RP problem as follows:

$$\min \sum_{x,y} (1 - d_{xy}) t_{xy} \quad (2a)$$

$$\text{s.t. } d_{xy} + d_{xz} + d_{yz} \leq 2, \forall 1 \leq x < y < z \leq |V_A| \quad (2b)$$

$$d_{xy} - d_{xz} - d_{yz} \leq 0, \forall 1 \leq x < y \leq |V_A|, z \neq x, y \quad (2c)$$

$$d_{xy} \in \{0, 1\} \forall x, y \in |V_A|. \quad (2d)$$

Equation (2a) represents the objective function, minimizing total interference among vertices within the same partition. The cut is modeled by the triangle inequalities in (2b) and (2c). Each facet of the cut is induced by these inequalities, which involve three vertices. When a facet intersects the cut plane, two vertices are on one side, and the third is on the other, making d_{xy} zero for vertices on the same side. Equation (2b) ensures the sum of d_{xy} is less than or equal to two, while (2c) ensures d_{xy} is one for vertices on opposite sides. If a facet does not intersect the cut plane, d_{xy} is zero for all pairs, consistent with (2b) and (2c). To minimize path interference, facets are forced to intersect the cut whenever possible.

B. Phase 2: Analytical model for single-hop link

This section introduces the exact analytical CTMC model for SS-EONs considering defragmentation. By generating all

feasible states and transitions, the model accurately estimates BP for both single- and multi-class requests.

1) *State generation with defragmentation*: The BP in SS-EONs is determined using the states of an analytical model. State formation occurs as lightpath requests from different classes in K arrive and depart, taking into account cores and modes [35]. A defragmentation state, denoted by $v \in C_D$, represents a state where retuning is in progress, based on both serving and waiting requests. Here, serving requests refer to allocated existing requests in a state, while waiting requests represent new incoming requests awaiting retuning. The transition rates between a defragmentation state $v \in C_D$ and a normal state $u \in C$ are defined using the transition function $\alpha(u, v)$ as follows: (i) When defragmentation is completed, the transition rate from defragmentation state v to normal state $u \in C$ is given by $\mu_d/|C_v^+|$, if $u \in C_u^+ \cap C$. (ii) If a lightpath request completes its service time during the retuning process, it immediately enables a waiting request by finalizing the ongoing retuning process. In this case, the transition occurs with rate $r_k \times \mu_k/|C_u^{k,-}|$, if $u \in C_u^{k,-} \cap C$, where r_k represents the number of allocated requests for each class $k \in K$ in state u . Additionally, if $\alpha(u, v) = 0$, it indicates that state $u \in C$ is not a target state for defragmentation state $v \in C_D$.

Algorithm 1 Generating feasible states and their transitions for a link considering counter-propagation

Require: Sets of cores P , modes Q , classes K , fibers F , and partition B
Ensure: State transition diagram including normal states C and defragmentation states C_D

```

1: Initialize  $C$ ,  $C_D$ ,  $C_u^{pqk,+}$ ,  $C_u^{pqk,-}$ ,  $L_k$ , and  $J_k$  as empty sets for all  $p \in P$ ,  $q \in Q$ ,  $k \in K$ , and  $u \in C$ 
2: Initialize state 1 as empty and assign it to  $C$ , arranged by fiber partitions
3: Set  $u = 1$ 
4: while  $u \leq |C|$  do
5:   for  $k = 1$  to  $|K|$  do
6:     if request in class  $k$  uses forward or backward fiber link then
7:       Call  $\text{States\_in\_Forward}$  (backward) ( $C$ ,  $C_u^{pqk,+}$ ,  $C_u^{pqk,-}$ )
8:     if  $C_u^{pqk,+} \not\subseteq C$  then
9:        $C \leftarrow C \cup C_u^{pqk,+}$ 
10:    for  $k = 1$  to  $|K|$  do
11:      After departure of a class  $k$  request from state  $u$  using mode  $q$  of core  $p$ , enqueue resulting states into  $C_u^{pqk,-}$ 
12:      if  $C_u^{pqk,-} \not\subseteq C$  then
13:         $C \leftarrow C \cup C_u^{pqk,-}$ 
14:       $u \leftarrow u + 1$ 
15: {Defragmentation state generation from normal states}
16: Set  $v = 1$ 
17: for  $v = 1$  to  $|C|$  do
18:   for  $k = 1$  to  $|K|$  do
19:     if available slots in state  $v$  are sufficient for a class  $k$  request then
20:       if slots are not contiguous then
21:         if defragmentation state for  $v$  is not in  $C_D$  then
22:           if allocation uses non-interfering core-mode pair then
23:             Apply parallel defragmentation; create new state  $v$ 
24:              $C_D \leftarrow C_D \cup \{v\}$ ,  $C_v^{pqk,+} \leftarrow C_v^{pqk,+} \cup \{v\}$ 
25:           else
26:             Apply single lightpath defragmentation; create new state  $v$ 
27:              $C_D \leftarrow C_D \cup \{v\}$ ,  $C_v^{pqk,+} \leftarrow C_v^{pqk,+} \cup \{v\}$ 
28:              $v \leftarrow v + 1$ 
29:           else
30:             Continue
31:         else
32:           Mark state  $v$  as blocking for class  $k$ :  $L_k \leftarrow L_k \cup \{v\}$  (or  $J_k \leftarrow J_k \cup \{v\}$ )

```

```

32: {Defragmentation state generation from existing defragmentation states}
33: Set  $v = 1$ 
34: while  $v \leq |C_D|$  do
35:   After defragmentation completes, transitions to normal states are assigned to  $C_v^+$ 
36:   for  $k = 1$  to  $|K|$  do
37:     if defragmentation state has enough resources for a new class  $k$  request and its waiting request then
38:       if a new defragmentation state  $v'$  is not in  $C_D$  then
39:         Add  $v'$  to  $C_D$ 
40:       else
41:         Mark  $v$  as blocking:  $L_k^D \leftarrow L_k^D \cup \{v\}$  (or  $J_k^D \leftarrow J_k^D \cup \{v\}$ )
42:       After a service completes during retuning, transition to a normal state is added to  $C_v^{k,-}$ 
43:        $v \leftarrow v + 1$ 

```

States_in_Forward (backward) (C , $C_u^{pqk,+}$, $C_u^{pqk,-}$)

```

1: Check required slots for class  $k \in K$  in partition 0 (partition 1) of the forward fiber and in partition 1 (partition 0) of the backward fiber.
2: if slots are available in partition 0 (partition 1) of the forward fiber then
3:   if interference exists among selected core-mode combinations then
4:     Enqueue all feasible states into  $C_u^{pqk,+}$  after arrival of a class  $k$  request at state  $u$ , using mode  $q \in Q$  of core  $p \in P$ , ensuring XT-avoidance constraints are satisfied.
5:   else
6:     Enqueue all feasible states into  $C_u^{pqk,+}$  after arrival of a class  $k$  request at state  $u$ , using mode  $q$  of core  $p$  in partition 0 (partition 1) of the forward fiber.
7: else if slots are available in partition 1 (partition 0) of the backward fiber then
8:   if interference exists among selected core-mode combinations then
9:     Enqueue all feasible states into  $C_u^{pqk,+}$  after arrival of a class  $k$  request at state  $u$ , using mode  $q$  of core  $p$  in partition 1 (partition 0) of the backward fiber, satisfying XT-avoidance constraints.
10:   else
11:     Enqueue all feasible states into  $C_u^{pqk,+}$  after arrival of a class  $k$  request at state  $u$ , using mode  $q$  of core  $p$  in partition 1 (partition 0) of the backward fiber.
12: else
13:   State  $u$  is marked as a blocking state for class  $k \in K$  in the forward (backward) fiber.
14:    $L_k = L_k \cup \{u\}$  (or  $J_k = J_k \cup \{u\}$ ).

```

Algorithm 1 generates all feasible states and their transitions for a link considering defragmentation. It begins by initializing the sets C , C_D , $C_u^{pqk,+}$, $C_u^{pqk,-}$, L_k , and J_k as empty (line 1). In line 2, the initial state ($u = 1$) is constructed using both forward and backward fiber links, each divided into partitions 0 and 1. This state, where all spectrum slots are free, is then added to set C . Line 3 initializes the state index variable $u = 1$. Lines 5–9 handle request allocations for class $k \in K$ in state u , exploring all valid combinations of core $p \in P$, mode $q \in Q$, and spectrum slots. Line 6 determines whether the request is routed via the forward or backward fiber link. Based on this, the respective subroutine, $\text{States_for_Forward}$ or $\text{States_for_Backward}$, is invoked in line 7. If the resulting state $C_u^{pqk,+}$ is not already in C , it is added (line 9). Line 11 considers request departures from class k in state u , freeing resources on mode q and core p . The resulting state $C_u^{pqk,-}$ is added to C in line 13 if not already present. This process iterates across all combinations until the complete normal state space is generated (lines 4–14). Defragmentation begins with line 15, where a new integer variable v is initialized. Lines 18–19 check whether available spectrum slots for a class- k request in state u are non-contiguous. If yes, line 20

verifies whether a corresponding defragmentation state already exists in C_D . Line 21 checks whether the lightpath is assigned to a core-mode pair with a non-interference link (Fig. 2(d)). If true, parallel defragmentation for multiple lightpaths is applied in line 23, and the new state v is added to C_D and $C_v^{pqk,+}$. If the allocation is on an interference core-mode pair, defragmentation is limited to a single lightpath, and the resulting state is similarly added in line 26. A separate defragmentation transition process is initiated at line 32 by resetting $v = 1$. Line 34 models the completion of the retuning process, after which the state transitions back to normal and is assigned to set C_v^+ . Lines 36–38 handle the arrival of new requests within ongoing defragmentation states. If adequate slots are available and all constraints are met, a new defragmentation state is generated. If it does not already exist in C_D (line 37), it is added in line 38. If the condition in line 36 is not met, line 39 classifies the state as blocking, and it is added to the set L_k^D in line 40. Finally, line 41 addresses service completion during retuning. In such cases, the state transitions directly to a normal state by accepting pending waiting requests and is added to the set $C_v^{k,-}$.

In DAAM-CP, set $B_{p,q}^i$ represents the element of partition $i \in \{1, 2\}$ in set of fiber $F \in \{1, 2\}$, where each element where each element contains the combination of core $p \in P$ and mode $q \in Q$. Each feasible state in C consists of $|P||Q||R||F|$ spectrum slots, denoted by $s_{r,b_{p,q}}^f$, where $r \in R$, $b_{p,q} \in B_{p,q}^i$, $i \in \{1, 2\}$, and $f \in F$. Each feasible state is represented by: $(s_{1,b_{1,1}}^1, s_{2,b_{1,1}}^1, \dots, s_{|R|,b_{1,1}}^1), (s_{1,b_{2,1}}^1, s_{2,b_{2,1}}^1, \dots, s_{|R|,b_{2,1}}^1), \dots, (s_{1,b_{|P|,1}}^1, s_{2,b_{|P|,1}}^1, \dots, s_{|R|,b_{|P|,1}}^1), \dots, (s_{1,b_{|P|,|Q|}}^1, s_{2,b_{|P|,|Q|}}^1, \dots, s_{|R|,b_{|P|,|Q|}}^1), (s_{1,b_{2,1}}^2, s_{2,b_{2,1}}^2, \dots, s_{|R|,b_{2,1}}^2), (s_{1,b_{2,1}}^2, s_{2,b_{2,1}}^2, \dots, s_{|R|,b_{2,1}}^2), \dots, (s_{1,b_{|P|,1}}^2, s_{2,b_{|P|,1}}^2, \dots, s_{|R|,b_{|P|,1}}^2), \dots, (s_{1,b_{|P|,|Q|}}^2, s_{2,b_{|P|,|Q|}}^2, \dots, s_{|R|,b_{|P|,|Q|}}^2), (s_{1,b_{1,1}}^2, s_{2,b_{1,1}}^2, \dots, s_{|R|,b_{1,1}}^2), (s_{1,b_{2,1}}^2, s_{2,b_{2,1}}^2, \dots, s_{|R|,b_{2,1}}^2), \dots, (s_{1,b_{|P|,1}}^2, s_{2,b_{|P|,1}}^2, \dots, s_{|R|,b_{|P|,1}}^2), \dots, (s_{1,b_{|P|,|Q|}}^2, s_{2,b_{|P|,|Q|}}^2, \dots, s_{|R|,b_{|P|,|Q|}}^2), (s_{1,b_{1,1}}^2, s_{2,b_{1,1}}^2, \dots, s_{|R|,b_{1,1}}^2), (s_{1,b_{2,1}}^2, s_{2,b_{2,1}}^2, \dots, s_{|R|,b_{2,1}}^2), \dots, (s_{1,b_{|P|,1}}^2, s_{2,b_{|P|,1}}^2, \dots, s_{|R|,b_{|P|,1}}^2), \dots, (s_{1,b_{|P|,|Q|}}^2, s_{2,b_{|P|,|Q|}}^2, \dots, s_{|R|,b_{|P|,|Q|}}^2), where $s_{r,b_{p,q}}^f = 0$ denotes available slot with $p \in P$, $q \in Q$, $r \in R$, $f \in F$, and $b_{p,q}^i \in B_{p,q}^i$. $s_{r,b_{p,q}}^f = k$ indicates that the slot is utilized by a request with $p \in P$, $q \in Q$, $r \in R$, $f \in F$, and $b_{p,q}^i \in B_{p,q}^i$. $s_{r,b_{p,q}}^f = x$ indicates that the slot is used for avoiding IC-XT and IM-XT with $p \in P$, $q \in Q$, $r \in R$, $f \in F$, and $b_{p,q}^i \in B_{p,q}^i$.$

2) *Example of model*: Figure 5 illustrates the state transitions under the CMS-FF allocation policy. Due to space limitations, only a subset of states is shown. We consider $|P| = 3$, $|Q| = 2$, $|R| = 2$, $|K| = 1$, and $z_1 = 2$; adjacency of cores and modes are depicts in Fig. 2(b). DAAM-CP adopts counter-propagation for bidirectional lightpath allocation. Each class-1 request occupies two spectrum slots. Each fiber is partitioned into two disjoint sets (partitions 0 and 1) using a bi-partitioning approach that combines cores and modes.

States are classified into two types: normal states, denoted by u , and defragmentation states, denoted by v . A state is represented using 12 vectors of 5-slot values each, such as: $(s_{1,b_{1,2}}^1, s_{2,b_{1,2}}^1, \dots, s_{5,b_{1,2}}^1), (s_{1,b_{2,1}}^1, \dots, s_{5,b_{2,1}}^1), (s_{1,b_{3,2}}^1, \dots, s_{5,b_{3,2}}^1)$, and so on up to the 12th vector. The state generation process begins from the initial normal state $u = 1$, where all vectors have slot values (0,0,0,0,0), indicating all slots are available. Here, 0 denotes a free slot, 1 denotes an allocated slot, and x represents an XT-avoided slot. State transitions are generated using Algorithm 1. In Fig. 5, normal state $u = 125$ cannot accommodate a new class-1 request due to spectrum fragmentation, despite sufficient slots being available. To address this, defragmentation state $v = 1$ is activated, initiating a lightpath retuning process. This state may include vectors like $(\gamma, \gamma, \gamma, 0, 0), (0, 1, 1, 0, 0), \dots, (0, 1, 1, 0, 0)$ or $(0, 1, 1, 0, 0), (\gamma, \gamma, \gamma, 0, 0), \dots, (0, 1, 1, 0, 0)$ or $(0, 0, \gamma, \gamma, \gamma), (0, 1, 1, 0, 0), \dots, (0, 1, 1, 0, 0)$ or $(0, 1, 1, 0, 0), (0, 0, \gamma, \gamma, \gamma), \dots, (0, 1, 1, 0, 0)$, where γ represents slots undergoing retuning, and $x\gamma$ indicates XT-avoided slots during defragmentation. A total of 20 such sub-states are included in $v = 1$, each with a unique retuned lightpath. From state $u = 125$, 12 defragmentation states are generated under CMS-FF, each with an arrival rate of $\lambda_1/12$. Once retuning and allocation are completed, transitions return to normal states at a departure rate of μ_d . Overall, the algorithm generates $|u| = 524,785$ normal states and $|v| = 144$ defragmentation states. Defragmentation states are created in two cases: (i) when new requests cannot be allocated due to fragmented but sufficient slots, and (ii) when additional requests arrive in ongoing defragmentation states. States $v = 1, 2, \dots, 12$ emerge from $u = 125$ via parallel defragmentation on slots 2 and 3 of mode 1 in core 1 of fiber 1. The parallel defragmentation strategy allows simultaneous retuning of multiple requests, resulting in states $v = 2, 3, \dots, 12$. These, in turn, may trigger further defragmentation states. For instance, state $v = 211$ emerges from $v = 1$ after a lightpath completes service, and state $v = 237$ is generated due to a new request arrival post-retuning.

3) *Model analysis*: This section details and analyzes DAAM-CP. We formulate equilibrium equations of normal state u for a link in SS-EONs as follows:

$$\left(\sum_{f \in F} \sum_{p \in P} \sum_{q \in Q} \sum_{\substack{k \in K \\ C_{pqk}^{f,+} \neq 0}} \lambda_{pqk}^f + \sum_{f \in F} \sum_{p \in P} \sum_{q \in Q} \sum_{\substack{k \in K \\ C_{pqk}^{f,-} \neq 0}} \iota(u) \mu_{pqk}^f \right) A_u = \sum_{\substack{j \in C \\ u \neq j}} \left(\sum_{f \in F} \sum_{p \in P} \sum_{q \in Q} \sum_{\substack{k \in K \\ C_{pqk}^{f,+} \neq 0}} \lambda_{pqk}^f + \sum_{f \in F} \sum_{p \in P} \sum_{q \in Q} \sum_{\substack{k \in K \\ C_{pqk}^{f,-} \neq 0}} \mu_{pqk}^f \right) A_j + \sum_{v \in C_D} \alpha(u, v) B_v, \quad \forall u \in C. \quad (3)$$

In (3), λ_{pqk}^f represents the transition rate into state u caused by the arrival of a request in class $k \in K$ using mode $q \in Q$ of core $p \in P$ over either forward or backward fiber link $f \in F$. Conversely, μ_{pqk}^f denotes the rate of transitions leaving state u due to the departure of such a request. The total number of outgoing transitions from state u is denoted by $\iota(u)$, accounting for all departing requests in class k that use any direction of the fiber link. The probability that the system is in normal state u is denoted by A_u . State transitions occur due to

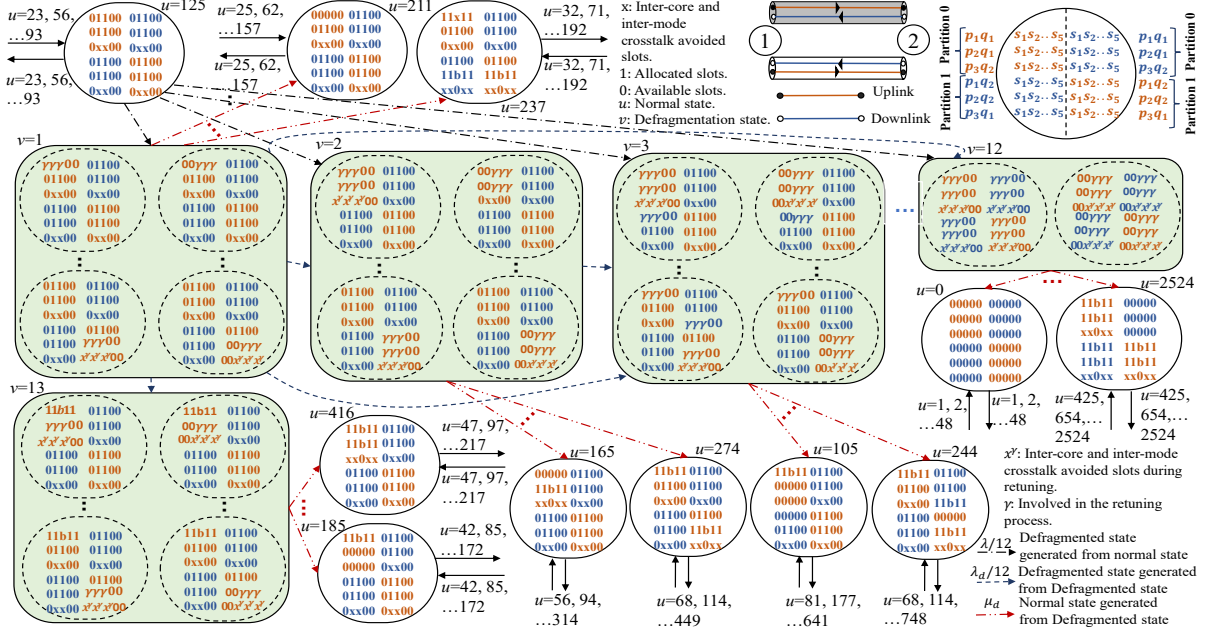


Fig. 5: Partial state transition diagram under CMS-FF policy with $|P| = 3$, $|Q| = 2$, $|R| = 5$, and $|K| = 1$. Here, 0 = available slot, 1 = allocated slot, x = XT-avoided slot (due to IC-XT/IM-XT), γ = retuning slots, x' = XT-avoided slots during retuning, u = normal state, v = defragmentation state.

the arrival or departure of requests in class k , utilizing mode q of core p over link f in either direction. In (3), the left-hand side (LHS) captures the total rate at which transitions occur out of state u , weighted by the state probability A_u . The first term on the LHS corresponds to the transition rate due to incoming requests in class k arriving at state u (when $C_u^{pqkf,+}$ is non-empty). The second term on the LHS captures the rate of transitions out of state u due to the departure of requests (when $C_u^{pqkf,-}$ is non-empty). The right-hand side (RHS) of (3) represents all possible transitions into state u . The first RHS term accounts for arrivals into u from another state j —specifically, when a request in class k (using mode q of core p and fiber f) enters u . The second RHS term captures the transitions into state u caused by a departure from another state j , again due to a request in class k utilizing mode q , core p , and fiber f . Finally, the last term on the RHS reflects transitions from any defragmentation state C_D to normal state $u \in C$, completing the description of how state u is reached through different transition events in the system.

The equilibrium equations of defragmentation state $v \in C_D$ can be given as:

$$\left(\sum_{j \in F} \sum_{p \in P} \sum_{q \in Q} \sum_{k \in K} \lambda_{pqk}^j + \sum_{j \in F} \sum_{p \in P} \sum_{q \in Q} \sum_{k \in K} z_k \mu_{pqk}^j + \mu_d \right) B_v = \sum_{u \in C} \sum_{j \in F} \sum_{p \in P} \sum_{q \in Q} \sum_{k \in K} \lambda_{pqk}^j A_u + \sum_{h \in C_D \setminus \{u\}} \sum_{p \in P} \sum_{q \in Q} \sum_{k \in K} \lambda_{pqk}^j B_h \quad \forall v \in C_D. \quad (4)$$

In (4), the first term on the LHS represents the transition rate out of defragmentation state u due to the arrival of a request in class $k \in K$, using mode $q \in Q$ of core $p \in P$ over either

forward or backward fiber link $f \in F$. The second LHS term accounts for transitions from state u due to the departure of all existing requests (z_k) in class k using the same core-mode pair and fiber direction. The third term captures transitions resulting from the completion of the retuning process during defragmentation, governed by rate μ_d . The first term on the RHS indicates the transition rate into defragmentation state u from any state where an arriving request in class k over link f triggers defragmentation, leading to state $v \in C_D^{pqk,+}$. The second RHS term captures transitions from other defragmentation states $h \in C_D \setminus u$ that also lead to $v \in C_D^{pqk,+}$ due to new arrivals. For instance, the equilibrium equation for defragmentation state $v = 1$ in Fig. 5 using (4) is: $(24\mu_1 + \mu_d + \lambda_1)B_1 = \lambda_1 A_{125}$.

To compute BP under any spectrum allocation policy, equilibrium equations are applied to all system states. The resulting stationary distribution must satisfy the normalization condition: $\sum_{u \in C} A_u + \sum_{v \in C_D} B_v = 1$. By solving the set of equations derived from (3), (4), and the normalization constraint, we obtain A_u for all normal states $u \in C$ and B_v for all defragmentation states $v \in C_D$. The overall BP is then given by the sum of probabilities for all blocking states: $\sum_{u \in L_k} A_u + \sum_{v \in J_k} B_v$.

C. Iterative approximate model

The DAAM-CP model described earlier in Section IV-B offers an exact evaluation of BP; however, it struggles with scalability when it is applied to links with a large number of cores, modes, and spectrum slots. To address this limitation, we introduce an iterative approximation model for estimating